

# Syarfandi "Andi" Achmad

Backend Developer | API Architecture & Database Scaling Specialist Makassar, Indonesia | +62 852-5652-2335 | syarfandi.achmad@gmail.com | Portfolio: syarfandi.github.io | LinkedIn

## Professional Summary

Reliable **Backend Developer** with 4+ years of experience in building scalable, high-performance server-side architectures. Expert in **Node.js**, **API Design**, and the **PostgreSQL/Supabase** ecosystem. Proven track record of designing complex database schemas, optimizing query performance for high-volume transactions, and ensuring 99.9% system uptime for enterprise and government platforms. Dedicated to writing clean, maintainable code and architecting data-intensive applications for production-grade scale.

## Core Technical Competencies

- Backend Frameworks:** Node.js, Express.js, Next.js (API Routes), Fastify, PM2 Process Management.
- API Design:** RESTful API Architecture, GraphQL, WebSocket (Real-time), JWT & OAuth Authentication, Rate Limiting & Throttling.
- Database & Data:** PostgreSQL (Stored Procedures, Indexing, Partitioning), Supabase (Realtime, RLS, Edge Functions), Redis Caching, Database Migration & Versioning.
- DevOps & CI/CD:** Docker, GitHub Actions, Automated Testing, PM2, Linux Server Administration, Nginx Reverse Proxy.
- Cloud & Serverless:** Vercel Serverless Functions, Supabase Edge Functions, Cloud-native Architecture, Background Job Queues.
- Soft Skills:** Asynchronous Collaboration, Technical Documentation, Remote Team Leadership, System Design.

## Technical Projects

- Kawan Sedarah (Blood Stock):** A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- Portal Data Susel:** A unified information system delivering **data visualization**, **monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- AI Virtual Assistant:** A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- GetKasir Enterprise POS:** A large-scale **Enterprise** POS ecosystem with **100+ modules**, integrating **AI-driven** sales prediction, **biometric** authentication, and a scalable **cloud-native** architecture.
- PPID Kemenag Kota Parepare:** A public information disclosure portal for **PPID Kemenag** providing access to **official documents**, **performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- OP Tracker:** A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter**, **saga & arc** navigation map, and a gamified **pirate rank** system.
- QR Scanner Presence App:** A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js** + **Supabase** stack with **real-time** sync.
- Malmora (Reseller & Dropship):** A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi and Eastern Indonesia**.
- Email Server Makassar Gov:** A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- Helpdesk Kota Makassar:** An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration across **Makassar City agencies**.
- Virtual Museum Makassar:** An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural heritage.
- Al-Haram VR:** A high-precision **virtual reality** simulation of **Al-Masjid Al-Haram** and **Al-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrah educational tool.
- Paralluta (Lalu Lintas):** A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
- SOP Pelaporan Insiden Siber:** A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.

## Professional Experience

### Computer Specialist (Civil Servant) | Ministry of Religious Affairs (Regional Office)

2025 – Present | Makassar, Indonesia

- Engineered** RESTful APIs and backend services for government platforms using **Node.js** and **Supabase**, serving 9M+ records with sub-200ms response times.
- Architected** the **Data Susel** data warehouse with PostgreSQL stored procedures, aggregation pipelines, and real-time analytics dashboards.
- Implemented** JWT-based authentication, row-level security (RLS), and automated database migrations for multi-tenant government services.
- Automated** internal reporting workflows through custom backend scripts, reducing manual data processing time by 60%.

### DevOps & Infrastructure Expert | Department of Communication & Informatics (Diskominfo)

2023 – 2025 | Makassar, Indonesia

- Managed** PostgreSQL databases across 200+ on-premise server environments, implementing automated backup strategies, query optimization, and connection pooling.
- Developed** a high-concurrency **Mailcow** mail cluster backend handling 200+ government domains with automated provisioning scripts and DNS API integrations.
- Designed** RESTful APIs for the **Helpdesk Makassar** citizen complaint platform, supporting ticket lifecycle management and cross-department routing.

### Co-Founder | GetKasir (Enterprise POS Startup)

2019 – Present | Remote/Project-based

- Designed** and built the entire backend architecture for a **100+ module enterprise POS system** using Node.js and PostgreSQL, handling thousands of daily transactions.
- Optimized** complex SQL queries, indexing strategies, and database partitioning, reducing API response times by 40% under high-volume loads.
- Implemented** Supabase Realtime subscriptions, webhook integrations, and background job queues (Bull/Redis) for merchant dashboard synchronization.
- Built** inventory management APIs with batch processing, stock reconciliation logic, and multi-branch data synchronization.

### SQA & DevOps Engineer | Malmora (Startup)

2020 – 2023 | Remote

- Built** the backend API layer for a reseller & dropship e-commerce platform serving Sulawesi and Eastern Indonesia markets.
- Implemented** order processing pipelines, inventory synchronization services, and automated logistics integration with third-party courier APIs.
- Designed** normalized database schemas for multi-vendor product catalogs, user role management, and commission tracking systems.
- Automated** CI/CD pipelines and implemented API testing suites, reducing deployment failures by 35%.

### Secretary | Nurul Huda Mosque

2021 – 2026 | Makassar, Indonesia

- Managed** organizational administration and coordinated the 2021-2026 board of directors.
- Digitized** community asset management through custom database solutions and automated reporting tools.

## Certifications & Awards

- National Cyber Exercise #10** – BSSN (2024)
- Top 10 Winner** – Samsung Indonesia Next Apps 4.0 Gear VR Challenge (2017)
- Microsoft CityApp Appathon** – Participant & Developer

## Education

- Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*

## Backend Expertise Keywords

Node.js, Express.js, Next.js API Routes, PostgreSQL, Supabase, REST API, GraphQL, WebSocket, JWT Authentication, OAuth, Redis, Docker, GitHub Actions, CI/CD, Database Optimization, SQL Query Tuning, Indexing, Database Migration, Row-Level Security (RLS), Serverless Functions, Edge Functions, Background Jobs, Bull Queue, PM2, Nginx, Linux, Microservices, API Gateway, Webhook Integration, Real-time Data Streaming, Multi-tenant Architecture, Database Partitioning, Connection Pooling.

## Future-Ready Technical Roadmap

To stay competitive in the global Backend market, I am actively implementing:

- Distributed Systems:** Designing highly-available, event-driven architectures with Kafka and RabbitMQ.
- AI-Driven APIs:** Integrating AI models directly into backend pipelines for real-time intelligence.
- Vector Databases:** Implementing Pinecone/Milvus for efficient AI similarity searches and RAG systems.
- Edge Databases:** Utilizing distributed databases like Turso/Supabase Edge for low-latency global data access.